Literature-Implied Answer for arXiv:1607.08488

fa_banach_001, agent_lane_01

2026-07-03

Abstract

Sain's arXiv:1607.08488 conjectured that on ℓ_p^n, $1 < p < \infty$, and more generally on every finite dimensional strictly convex smooth real Banach space, a left symmetric operator for Birkhoff-James orthogonality must be the zero operator. Paul, Mal, and Wojcik later proved a stronger compact operator theorem. In finite dimension it applies directly, so both conjectures are affirmatively settled by later literature.

Source conjectures

The source paper proves the zero-operator-only result for ℓ_p^2 when $p \geq 2$, and then states two conjectures:

$$\text{Conjecture 2.13:} \quad X = \ell_p^n,\ 1 < p < \infty, \quad T \in \mathcal{L}(X) \text{ left symmetric} \iff T = 0,$$

and

$$\begin{aligned} \text{Conjecture 2.14:} \quad & X \text{ finite dimensional, strictly convex, smooth, real,} \\ & T \in \mathcal{L}(X) \text{ left symmetric} \iff T = 0. \end{aligned}$$

The source also notes that Conjecture 2.14 implies Conjecture 2.13.

Supporting theorem

Paul, Mal, and Wojcik prove the following stronger statement in arXiv:1807.11166:

$$\text{if } X \text{ is reflexive and strictly convex, } Y \text{ is strictly convex, and } T \in K(X, Y),$$

then T is left symmetric if and only if $T = 0$.

Implication

Let X be as in Sain's Conjecture 2.14 and take $Y = X$. Since X is finite dimensional, it is reflexive and every bounded operator on X is compact. Since X is strictly convex, the range hypothesis Y strictly convex is also satisfied. Therefore the Paul–Mal–Wojcik theorem applies to every $T \in \mathcal{L}(X) = K(X, X)$ and gives

$$T \text{ left symmetric} \iff T = 0.$$

This is exactly the conclusion of Conjecture 2.14. Since ℓ_p^n, $1 < p < \infty$, is finite dimensional and strictly convex, the same theorem also proves Conjecture 2.13.

Status

The answer is affirmative by later literature. This packet is classified as a literature-implied answer because the later theorem subsumes the conjectures rather than explicitly naming them.

References

[1] Debmalya Sain. *Birkhoff-James orthogonality of linear operators on finite dimensional Banach spaces*. arXiv:1607.08488; J. Math. Anal. Appl. 447 (2017), 860–866.

[2] Kallol Paul, Arpita Mal, and Pawel Wojcik. *Symmetry of Birkhoff-James orthogonality of operators defined between infinite dimensional Banach spaces*. arXiv:1807.11166.